1. Use Part 1 of the Fundamental Theorem of Calculus to find the derivative of each of the following functions.

1). $g(x) = \displaystyle\int_1^x \frac{1}{t^4+2}\,dt.$ 2). $g(x) = \displaystyle\int_{x^2}^1 \cos\sqrt{t}\,dt.$ 3). $g(x) = \displaystyle\int_{-x}^{\cos x} (1+t^2)^5\,dt$

2. Evaluate the integral. (Note: Use the Fundamental Theorem of Calculus-Part 2)).

1). $\displaystyle\int_\pi^{2\pi} \tan\theta\cos\theta\,d\theta$ 2). $\displaystyle\int_1^9 \frac{x-1}{\sqrt{x}}\,dx$

3). $\displaystyle\int_{\pi/6}^{\pi/4} \sec^2 x\,dx$ 4). $\displaystyle\int_1^{\sqrt{3}} \frac{4}{t^2+1}\,dt$

5). $\int_{-1}^{2} (x^3 - 2x)\, dx$

6). $\int_{-1}^{1} e^{u+1}\, du$

7). $\int_{0}^{2} (2 + x^5)^2\, dx$

8). $\int_{0}^{2} 2^x\, dx$

3. Evaluate $\int_{-2}^{2} f(x)\, dx$, where $f(x) = \begin{cases} 3, & x \geq -1 \\ \dfrac{1}{x}, & x < -1 \end{cases}$.